

CS435 Introduction to Big Data

Syllabus

Course Sections: 001 and 801

Semester: Fall 2026

Lecture: 4:00-5:15PM, Monday and Wednesday, CSB130

Department of Computer Science, Colorado State University

1. Introduction to Big Data: Course Description

Modern scientific instruments and Internet-scale applications generate voluminous data pertaining to vital signs, weather phenomena, social networks that connect billions of users, and the origins of distant planets. Data produced in these settings hold the promise to significantly advance knowledge. This course covers fundamental concepts in Big Data. The course examines issues related to data analysis, organization, retrieval, and knowledge discovery at scale. This will include topics such as large-scale computing paradigms, petascale data storage systems, and scalable data analytics algorithms with their case studies. This course will involve hands-on programming assignments and a term project using real-world datasets.

1.1.Course Learning Objectives

Upon successful completion of this course students will be able to:

- Understand fundamental concepts for the big data
- Understand key computing paradigm and underlying scalable systems solutions
- Understand scalable algorithms for the big data analysis
- Formulate big data analysis problems and apply scalable algorithms alongside comprehensive analysis and evaluation
- Design, perform, evaluate, and conduct scientific analysis of results of distributed big data analysis over large-scale datasets

1.2.Course Topics

- Overview of Big Data analytics
- Scalable in-memory data analytics frameworks
- Scalable data analysis models (e.g., regression models, random forest, recommendation systems, distributed PageRank algorithm)
- Distributed storage architecture
- Fundamental concepts of NoSQL databases
- Big data analytics project using scalable computing frameworks and algorithms

1.3.Course Prerequisites

For undergraduate students:

Undergraduate level Operating Systems (CS370, C or higher) OR undergraduate level Algorithm Theory and Practice (CS320, C or higher) OR undergraduate level Machine Learning Foundations and Practice (CS345, C or higher)

For graduate students: There is no prerequisite for this course.

1.4.Lectures and Help Sessions

Class meeting: CSB130 (M, W 4:00PM – 5:15PM MST)

Professor: Sangmi Lee Pallickara, Ph.D.

Email: Sangmi.Pallickara@colostate.edu

Office: CSB 374

Office Hours: Fridays, 9:00–9:50 AM (MST), and by appointment

- Available in-person and online (via MS Teams)
- If this time slot does not work for you, please feel free to request an appointment via email.

Graduate Teaching Assistants

Freya LaLuna and Lucas Kitaev

GTA email address: CompSci_CS435@colostate.edu

Five Mandatory Lab sessions:

[Option 1] 11AM- noon on Mondays, CSB325

[Option 2] 5-6PM on Tuesdays, CSB325

[Option 3] Weekend online session for students in Section 801 [Schedule TBD]

Help Desk hours

Hybrid help desk hours are available during regular work hours.

- In-person location: CSB120
- Online: via MS Teams course page

Online-only help desk hours are available after work hours and on weekends.

Please see the Canvas course page for more details.

1.5.MYP (Meet Your Professor) Session for Online Students in 801 Section

During the first three weeks of this semester, online students in Section 801 will be invited to join a brief online meeting with the professor. This meet-and-greet is a great opportunity to connect, ask questions, and get to know your professor. A sign-up sheet will be available on the course Canvas page

1.6.Course Materials

Course Materials

All course materials will be available through Canvas except for interactive online course activities.

Interactive online activities (help sessions, online meetings, and discussions):

Please use the MS Teams course page.

Textbooks:

No textbook required.

Reading lists and online materials are provided through Canvas

2. Course Components and Grading

Course Components	Percentage	Notes
Programming Assignments (Individual)	20%	PA1 (5%) PA2 (5%) 5 Mandatory Lab sessions (Total 10% = 2% x 5)
Term Project (Group Activity)	20%	Proposal: 1% Final report: 8% Final presentation: 3% Discussion: 5% Team participation: 3% *Note: For the individual scores may be different within a team based on the participation.
Exams	55%	Midterm 25% Final exam: 30%
Quizzes (Individual)	5%	12-14 quizzes (3 lowest scores will be eliminated)
Extra credits	5%	Worksheets (Individual) In-class participation (Individual)

Participation Requirements

This course includes collaborative team activities. Your active participation is required. Your participation will be assessed by your peer team members and accounted for your final grade.

Make Up Policy

Exams: Makeup exams are only given in extraordinary circumstances. If you have a family or medical emergency that prevents you from attending the midterm, or the final, students must consult with the professor as soon as possible, preferably before the start of the exam. Students must provide a valid document to reschedule any course activities. Course examination dates are listed in the syllabus; be aware of them and plan accordingly. No make-ups will be given for the missed mandatory lab sessions and quizzes.

Letter Grading Policy

Letter grades will be based on the following standard breakpoints: ≥ 90 is an A, ≥ 89 is an A-, ≥ 88 is a B+, ≥ 80 is a B, ≥ 79 is a B-, ≥ 78 is a C+, ≥ 70 is a C, ≥ 60 is a D, and < 60 is an F. I will not cut higher than this, but I may cut lower.

Late Submission Policy

All of the assignments are to be submitted electronically using Canvas. Always check the assignment page for due dates. Assignments can be submitted up to a maximum of 2 days past the deadline. There will be a deduction of 10% (of the total score for the assignment) as the penalty per day. For example, if the assignment was due at 5:00 pm on Thursday: (1) you will lose 10% of the total score if you submit between 5:01 pm Thursday and 5:00 pm Friday, (2) if you submit between 5:01 pm Friday and 5:00pm Saturday, you will lose 20% of the points set aside for this assignment. (3) No submissions will be accepted after 5:00pm on Saturday and you will be given a 0 for that assignment.

Flex Tokens for Programming Assignments

You will start the semester with two Flex Tokens. A Flex Token waives the penalty of one day late submission. You can use these tokens either together or separately for two different programming assignments. The flex tokens are applicable **only** for the programming assignments (PA1 and 2).

If the submission deadline is September 1st and you submit your assignment on September 2nd with your 1 flex token, there will be no score deduction applied. These tokens cannot be reused. If you wish to use your token(s), you must inform the GTA when you submit the assignment. The GTAs will track the number of tokens you use. You are not allowed to apply any partial token.

3. Exams

Midterm

Section 001: 4:00PM--5:15PM, October 7, 2026 in class (CSB130)

Section 801/802: Online exam – a single 75-minute session between 4:00PM October 7, 2026 and 3:59PM October 9, 2026. Students may take the exam any time in this window.

Final exam

Section 001: 11:50AM -- 1:50PM, December 14, 2026, CSB130

Section 801/802: Online exam – a single 120-minute session between 11:50AM December 14, 2026 -- 11:49AM December 15, 2026 (Students may take the exam any time in this window. Students are given 120 minutes to complete the exam.)

For the CSU Final exam schedule: [[LINK](#)]

4. Weekly Schedule (Tentative, Subject to Minor Adjustment)

Please check the canvas page for the schedule every week.

Week	Topics	Key Deliverables	In-class Activities
UNIT 1: Introduction			
Week 1 8/24-30	Course Introduction and Overview of Big Data Introduction to In-memory Distributed Computing Framework	MYP session (online students)	Quiz/Worksheet/ Survey
UNIT 2: In-memory Distributed Computing Paradigm			
Week 2 8/31-9/6	Programming with RDDs	MYP session (online students) Lab #1- mandatory	Quiz/Worksheet
Week 3 9/7-13	Programming with RDDs Running on Cluster	MYP session (online students) 9/7: University Holiday – No Class Lab #2- mandatory	Quiz/Worksheet
Week 4 9/14-20	Spark SQL and Data Frames Apache Spark Design Patterns	Lab #3- mandatory	Quiz/Worksheet
UNIT 3: Scalable Analytics Algorithms			
Week 5 9/21-27	Web-Scale Link Analysis	Programing Assignment (PA) #1 submission (9/24)	Quiz/Worksheet
Week 6 9/28-10/4	Web-Scale Link Analysis Scalable clustering algorithms	PA #1 grading interview	Quiz/Worksheet
Week 7 10/5-11	Scalable clustering algorithms Midterm (10/7/2026) Online section: See the announcement	Midterm (10/7/2026) – in class Online section: See the announcement	Worksheet, midterm preparation review
Week 8 10/12-18	Recommendation systems	Lab #4- mandatory	Quiz/Worksheet
Week 9 10/19-25	Recommendation systems	Lab #5- mandatory	Quiz/Worksheet
Week 10 10/26-11/1	Scalable Machine Learning	PA #2 submission (10/29)	Quiz/Worksheet
Week 11 11/2-8	Scalable Machine Learning	PA #2 grading interview Final Term Project proposal submission (11/4)	Quiz/Worksheet
UNIT 4: Petascale Data Storage			
Week 12 11/9-15	Distributed File Systems	Final Term Project check-in #1 submission (11/12)	Quiz/Worksheet
Week 13 11/16-22	NoSQL Databases	Final Term Project check-in #2 submission (11/19)	Quiz/Worksheet

Week 14 11/23-29	Fall Break	11/21-11/29 Fall break	No Class
Week 15 11/30-12/6	NoSQL Databases	Final Term Project report and software submission (12/3)	Quiz/Worksheet
Week 16 12/7-11	Term Project Presentations	12/11 Last Day of classes Final Presentation	Students Presentations
Finals week	Final Exam (12/14, Monday) Online section: See the announcement	11:50AM – 1:50PM Online section: See the announcement	Final Exam

5. Course Policy

This course will adhere to the CSU Academic Integrity Policy as found on the Students' Responsibilities page [\[Link\]](#) of the CSU General Catalog and in the Student Conduct Code [\[Link\]](#).

5.1. Academic Integrity

We take academic integrity seriously. At minimum, academic integrity means that no one will use another's work as their own. The CSU writing center defines plagiarism this way:

"Plagiarism is the unauthorized or unacknowledged use of another person's academic or scholarly work. Done on purpose, it is cheating. Done accidentally, it is no less serious. Regardless of how it occurs, plagiarism is a theft of intellectual property and a violation of an ironclad rule demanding credit be given where credit is due."

If you plagiarize in your work you could lose credit for the plagiarized submission, fail the assignment, or fail the course. Plagiarism could result in expulsion from the university. Each instance of plagiarism, classroom cheating, and other types of academic dishonesty will be addressed according to the principles published in the CSU General Catalog (see page seven, column two: [\[Link\]](#)).

Of course, academic integrity means more than just avoiding plagiarism. It also involves doing your own reading and studying. It includes regular class attendance, careful consideration of all class materials, and engagement with the class and your fellow students. Academic integrity lies at the core of our common goal: to create an intellectually honest and rigorous community.

5.1. Use of Generative AI tools

In this course, students may use generative AI tools for out-of-class activities (e.g., programming assignments and term projects). However, AI does not assume authorship or accountability. Transparency is expected whenever AI makes a substantive contribution.

The use of generative AI tools and technologies to create content is permitted, but it must be fully disclosed. For example, students must acknowledge in writing that "ChatGPT" was used to generate text in the "Introduction and Methodology sections" of the final project report. Fabricating data, experiments, or citations, and generating fake analysis, are not permitted. Unacknowledged use of AI for creating content is a form of plagiarism.

5.2.COVID

Important information for students: All students are expected and required to report any COVID-19 symptoms to the university immediately, as well as exposures or positive tests (even home tests). If you suspect you have symptoms, or if you know you have been exposed to a positive person or have tested positive for COVID, you are required to fill out the COVID Reporter.

<https://covid.colostate.edu/reporter/>

If you know or believe you have been exposed, including living with someone known to be COVID positive, or are symptomatic, it is important for the health of yourself and others that you complete the online COVID Reporter. Do not ask your instructor to report for you.

If you do not have internet access to fill out the online COVID-19 Reporter, please call (970) 491-4600.

You may also report concerns in your academic or living spaces regarding COVID exposures through the COVID Reporter. You will not be penalized in any way for reporting.

When you complete the COVID Reporter for any reason, the CSU Public Health office is notified. Students who report symptoms or a positive antigen test through the COVID Reporter may be directed to get a PCR test through the CSU Health Network's medical services for students.

5.3.Inclusion

We create and nurture inclusive environments and welcome, value and affirm all members of our community, including their various identities, skills, ideas, talents and contributions.

This manifests in this class in our in-class discussions, discussion posts, and our interactions on Teams. You are to be kind to others with various backgrounds and identities, without making assumptions. With that said, this course is considered a 'Brave Space'. This does not exclude the classroom from being a Safe Space - and know you are welcome - and the classroom is meant to challenge ideals. We will often approach topics that may make you feel uncomfortable and challenge your point of view. We will even ask you to argue the opposite of what you believe. That is the joy of learning.

5.4.Copyright of the Course Materials

The course material of this course is the property of the instructor who developed this course and Colorado State University. Posting course materials on external sites (commercial or not) violates both copyright law and the CSU Student Conduct Code. Students who share content without instructor's permission could face legal action. If you maintain your code related to the course activities in a shared code repository (e.g., GitHub), ensure that your code remains private.